Mohamed Hossam

Backend Software Engineer

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SUMMARY

Software Engineer with Python, evaluating the trade-offs behind system performance, scalability, concurrency, data integrity and security. Competitive programming sharpens my edge-case reasoning, correctness and analytical thinking. Open to roles where I can apply this mindset to more challenging production problems.

TECHNICAL SKILLS

Backend & APIs: Django, DRF, FastAPI, REST APIs, Celery, WebSockets, Pydantic

Software Design & Quality: API Design, Service-Layer Architecture, Automated Testing

Databases: PostgreSQL, MySQL, Redis, SQLAlchemy, Database Indexing, Query Optimization.

Infrastructure & Deployment: Docker, Docker Compose, Nginx, Gunicorn, Uvicorn, Git, CI/CD

Languages: Python, SQL, JavaScript, C++, Bash

OPEN SOURCE CONTRIBUTIONS

NetBox — Merged PR #23378 *Oct 2026*

- Replaced NetBox global search's one-RQ-job-per-write indexing path with a Redis-backed coalescing buffer, merging updates across 98 searchable models into one drain job per 2-second window and collapsing a reported ~237K-job backlog.
- Made the pipeline concurrency-safe and crash-recoverable using atomic Redis batch claiming, PostgreSQL session-level advisory locking, a 5-minute recovery job, and bounded retries for poison batches; preserved eventual consistency and rolling-upgrade compatibility, added 67 tests.

NetBox — Merged PR #22834 *Jul 2026*

- Enforced context-aware HTML sanitization: restricted img[src] to HTTP(S)/relative via nh3 + regex; blocked ssh:/file:/ftp: bypasses across 3 rendering paths.
- Shipped 7 regression tests covering whitespace/control-char evasion, javascript: blocking, and scheme-intersection enforcement.

Django CRM — Merged PR #449 #456 *Jun–Aug 2026*

- Added Product.main_image with a database migration, image lifecycle cleanup, automatic 200×200 PNG resizing, and Django admin previews/thumbnails.
- Fixed TransactionQualityEvent admin UX: hid attachment icon when empty and added stacked inline with XSS-safe file links.

TECHNICAL PROJECTS

OCR Microservice *FastAPI, Tesseract, Docker, Pydantic v2, Prometheus* *2026 – Present*

- Engineered an OCR service to automate text extraction from images, turning unstructured visual content into machine-readable text for downstream processing.
- Benchmarked throughput at 7.2 requests/sec with 890 ms p95 latency under 50 concurrent requests; used Prometheus metrics and structured logs to analyze CPU utilization.
- Prevented overload by bounding CPU-bound Tesseract behind worker pool and queue backpressure with 429, selecting limits from measured throughput plateau and p99 tradeoffs.
- Reduced OCR error rate by 62% (CER 8.4% → 3.1%) via ablation-tested Pillow preprocessing, keeping only stages with proven accuracy gain vs latency cost.

YouTube Clone *Django 6, PostgreSQL 16, Redis 7, Celery 5, ClamAV, Docker, ImageKit CDN* *2026 – Present*

- Architected 3-layer platform with zero-byte app storage; served via signed ImageKit CDN URLs for adaptive HLS (240p–1080p).
- Orchestrated asynchronous Celery and ClamAV scanning with quarantine-first uploads, exponential-backoff retries and authorization gated on scan status; validated files using magic-byte sniffing, ffprobe and SHA-256 deduplication to reject malformed and duplicate.
- Concurrency-safe abuse controls: Redis token-bucket per IP; 24h view dedup with atomic F() increments and double-check transactions to prevent redundant writes under concurrent load.

LEADERSHIP & COMMUNITY

Leader, Mentor & Problem Setter *ICPC Nahda University Community* *2025 – Present*

- Mentored 400+ competitive programmers in algorithms, data structures.
- Authored internal training contests/practice **sheets** aligned with ICPC standards.

ACHIEVEMENTS

- Constructor Open Cup 2026: Ranked 20th internationally (Constructor University & JetBrains Foundation)
- QECPC 2025: Honorable Mention • ECPC Finalist 2026
- Codeforces: Candidate Master rating (top 45 in Egypt); solved 2,400+ algorithmic problems across Codeforces, LeetCode, and vJudge
- GitHub: Ranked #10 in Egypt by (**Committers.top**)

EDUCATION

B.Sc. in Computer Science Nahda University, Beni Suef *Expected 2028*
• GPA 3.26/4.0
Relevant Coursework: Algorithms, OOP, Database Systems, Operating Systems, Computer Networks